

CLOUD SECURITY ROADMAP

— BEGINNER TO JOB-READY GUIDE —

**LEARN.
PRACTICE.
PROTECT.
SUCCEED.**

YOUR STEP-BY-STEP
GUIDE TO BECOME
A CLOUD SECURITY
EXPERT



EASY TO
UNDERSTAND



STEP-BY-STEP
ROADMAP



REAL-WORLD
SECURITY SKILLS



CAREER
GROWTH



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LEARN CYBERSECURITY. DEFEND THE FUTURE.

1

What is Cloud Security?

Cloud Security means protecting

- Cloud Applications
- Cloud Data
- Cloud Servers
- Cloud Users

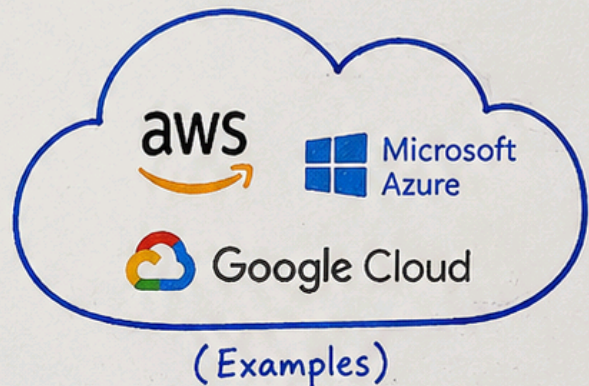
Simple Example:

When you save a file in Google Drive, it is protected by Cloud Security.



Why do companies use Cloud?

- It is cheap
- It is fast
- It is easy to use
- We can access from anywhere



Why do hackers attack the cloud?

- To steal data
- To lock data (Ransomware)
- To use resources for wrong things
- To break company systems



Hackers always look for weak security!



Cloud Security keeps our data, apps and users safe in the Cloud.

What is Cloud Computing?

Cloud Computing means using computers, storage, databases, software and other services over the internet instead of our own computer.

Think Simple :

Old Way

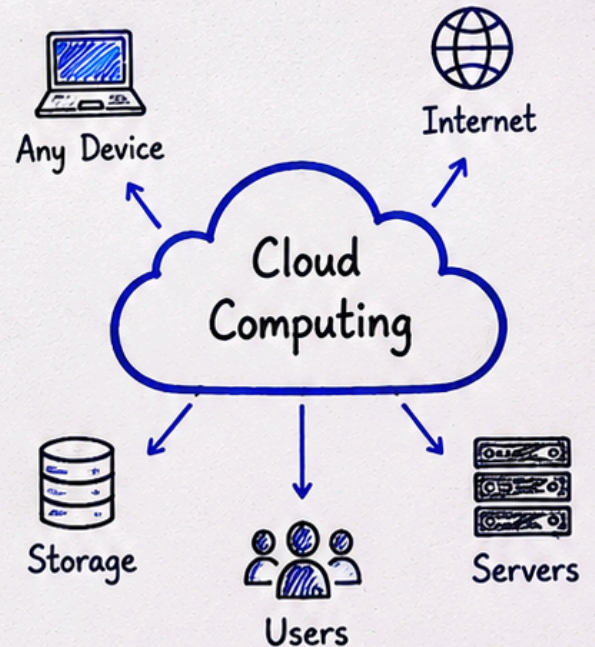


Everything in my computer

Cloud Way



Everything in the Cloud (Internet)



Big Cloud Providers



Amazon Web Services (AWS)



Microsoft Azure



Google Cloud Platform (GCP)

Simple Example :

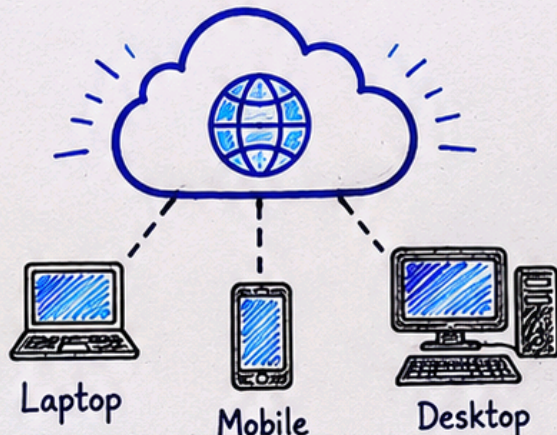
When you watch YouTube, the video is not in your phone.

It is in YouTube's cloud server.

That is Cloud Computing.

Benefits of Cloud Computing

- We can access data from anywhere.
- We don't need powerful computers.
- It saves money.
- It is fast and scalable.
- Companies can grow easily.



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How many types of Cloud exist?

There are
3 main types
of Cloud

Cloud can be used in different ways.
That is why we have 3 types.

1

Public Cloud

Cloud services are provided over the internet for **everyone**.



Example:

AWS, Microsoft Azure, Google Cloud

Simple Analogy:



Like a public bus. Anyone can use it.

2

Private Cloud

Cloud is used by only **one** organization.
It is **private** and secure.



Example:

A bank's private cloud, company's own cloud

Simple Analogy:

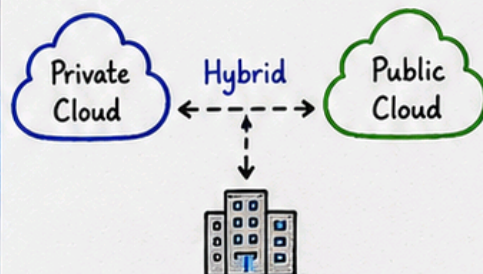


Like your own car. Only you can use it.

3

Hybrid Cloud

It is a combination of both **public** and **private** cloud.



Example:

Company keeps sensitive data in private cloud and other data in public cloud.

Simple Analogy:



Like using your car and public bus together.

Remember:

- ☑ Public Cloud = For everyone → public bus
- ☑ Private Cloud = For one organization → personal car
- ☑ Hybrid Cloud = Both together → bus + car



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Why is Cloud Security Important?



Cloud gives many benefits, but without security, it can also bring many **risks**.

Here's why Cloud Security is so important:

①



Hackers can steal data

Hackers try to steal company or user data stored in the cloud. This can cause money loss and loss of trust.

②



Password attacks

Weak passwords can be guessed or cracked by attackers. They can login and do bad things.

③



Ransomware attacks

Hackers lock your data and ask for money to unlock it. This can stop your business.

④



Data leaks

If settings are wrong, data can become public. Anyone on the internet can see it.

⑤



Account takeover

Attackers take control of user accounts and access everything. They can delete or change data.

Real World Example:



If a hacker steals hospital data (like patient records), it can harm patients, break trust and can be even life threatening.

Remember:

Cloud without security is like a house without a **lock**.



No Lock



With Lock



Good Cloud Security protects:



Data



Applications



Servers



Users



Business

Bottom line: Protect today, stay safe tomorrow. 😊

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Main Parts of Cloud Security

Many things work together to keep the cloud safe.

Here are the 6 main parts you must know!

1. IAM

Identity and Access Management.
Decides who can access what in the cloud.

Example:

Giving keys only to the right people.

2. MFA

Multi-Factor Authentication.
Adds extra security when logging in.

Example:

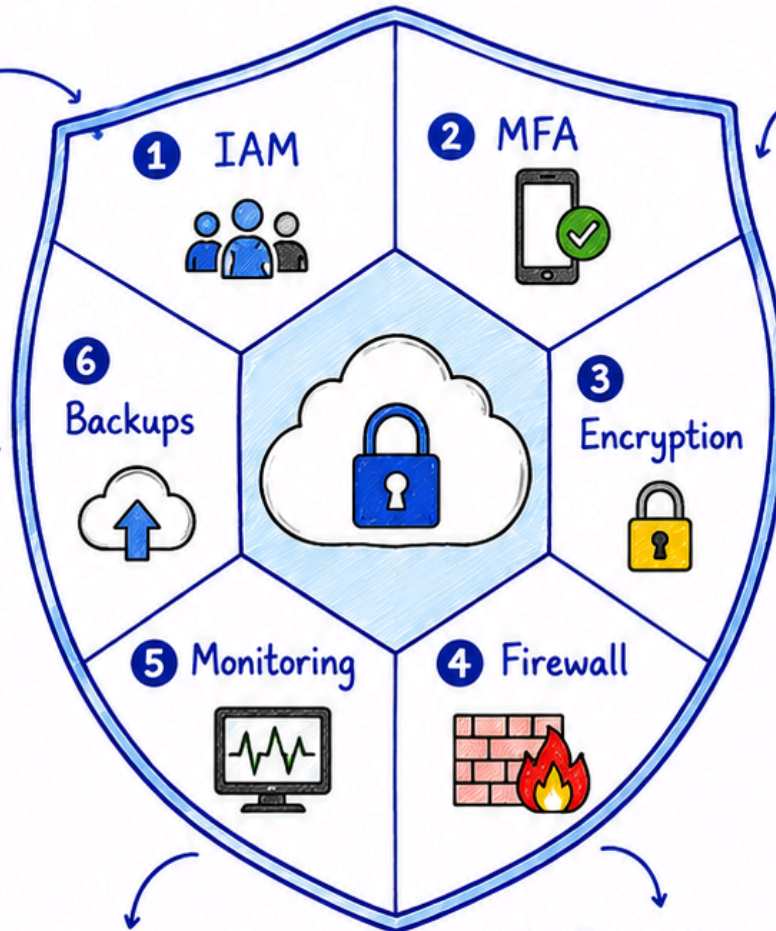
Password + Code from your phone.

6. Backups

Makes a copy of your data.
If something goes wrong, we can restore it.

Example:

Like saving your homework in a notebook copy.



3. Encryption

Converts data into a secret code.
Only the right key can open it.

Example:

Locking your diary with a secret lock.

5. Monitoring

Watches everything in the cloud.
Finds problems and unusual activity.

Example:

Like a CCTV camera that watches 24/7.

4. Firewall

Acts like a security guard.
It blocks bad traffic and allows safe traffic.

Example:

Only allowing visitors with ID inside the building.



Key Takeaway:

These 6 parts work together like a team.
If one part is weak, the whole security can break.
Strong cloud security keeps data, apps and users safe.



Remember:

Cloud security is not one tool.
It is a combination of many things working together.



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How do I start learning Cloud Security?



Simple Idea:

Learn step by step.
Build skills.
Practice.
Become job ready!

1 LEVEL 1 Computer Basics

- Understand computers
- Internet basics
- Files & folders
- Operating System
- Web browsers

Learn how computers and internet work first.

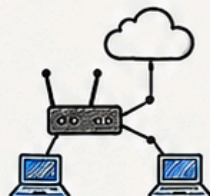


START

2 LEVEL 2 Networking Basics

- IP Address
- DNS
- Router & Switch
- HTTP / HTTPS
- Ports

Networking is like roads for internet traffic.



3 LEVEL 3 Linux Basics

- Linux commands
- File permissions
- Users & groups
- Logs and viewing

Most cloud servers use Linux.



4 LEVEL 4 Learn Cloud Platforms

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Cloud companies provide online computers and services.



5 LEVEL 5 Learn Security Basics

- IAM (Identity & Access)
- MFA (Multi Factor Auth)
- Encryption
- Firewalls & Security Groups
- Security Best Practices

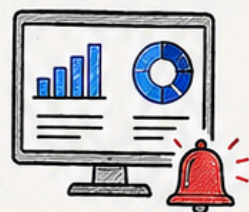
These are the building blocks of cloud security.



6 LEVEL 6 Learn Monitoring

- Logs
- SIEM Tools
- Alerts
- Threat Detection

Monitor everything and detect problems early.



7 LEVEL 7 Build Projects

- Secure an S3 Bucket
- Detect failed logins
- Create IAM users
- Enable Cloud Monitoring

Practice by building real small projects.



8 LEVEL 8 Get Certified

- AWS Cloud Practitioner
- Azure Fundamentals
- ISC2 Certified in Cybersecurity (CC)

Certifications show your knowledge to employers.



JOB READY



Small daily learning + Consistency + Practice = Big Success



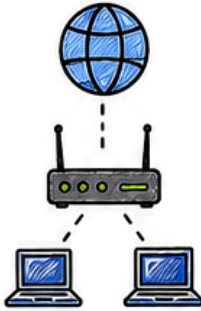
7

What skills do Cloud Security Engineers need?

① Networking

Understand how devices talk.

- IP Address
- DNS
- Ports
- VPN
- Firewalls



② Linux

Most cloud servers run on Linux.

- Commands
- Logs
- Permissions
- Users



③ Cloud Platforms

Learn at least one cloud platform well.

- AWS 
- Azure 
- Google Cloud 

Start with one, become strong.

④ IAM (Identity & Access Management)

Control who can access what in the cloud.

- Users
- Roles
- Permissions
- Least Privilege



⑤ SIEM & Monitoring

Monitor, detect and alert on threats.

- Splunk
- Wazuh
- Microsoft Sentinel
- Alerts & Dashboards



⑥ Incident Response

Know what to do when something goes wrong.



⑦ Scripting Basics

Automation saves time and reduces human mistakes.

- Python 
- Bash 

⑧ Communication

Explain security problems clearly and work with different teams.



Remember:

- ✓ You don't need to know everything at once.
- ✓ Focus on one skill at a time.
- ✓ Practice daily and be consistent.
- ✓ Real learning comes from doing projects.



You do NOT need to learn everything in one day.
Small steps every day lead to big success!



8 What does a Cloud Security Engineer do every day?



① Check Alerts

Security tools send warnings about problems.

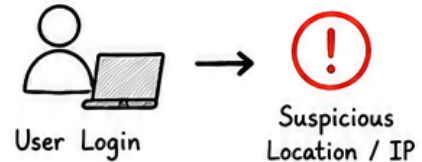
ALERTS 23		
TIME	SEVERITY	ALERT
10:15	High	Unauthorized login
10:10	Medium	EC2 instance open to IP
10:05	Low	Policy changed

MORNING



② Monitor Login Activity

Look for suspicious or unusual logins.



③ Fix Security Problems

Find and fix security issues.
Examples:

- Open ports
- Weak passwords
- Public cloud storage

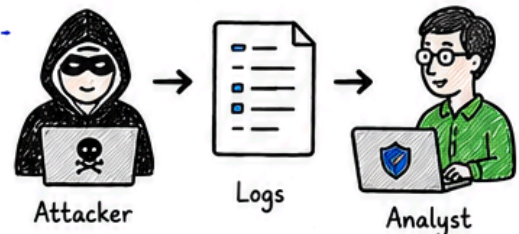


AFTERNOON



④ Investigate Attacks

Analyze logs and understand what happened.



⑤ Write Reports

Document issues, actions and recommendations.



NIGHT



⑥ Work with Teams

Collaborate with different teams to keep systems secure.



⑦ Use Security Tools

We use many tools every day.

- AWS Security Hub
- Microsoft Defender
- Wazuh
- CrowdStrike
- Splunk



splunk>

Some days are calm...



Some days feel like cyber firefighting!



Every day is different, but the goal is the same:

PROTECT the cloud. PROTECT the data. PROTECT the people.



9

How to become a Cloud Security Engineer?

≡ A Step-by-Step Career Roadmap ≡



Success Formula

- ✓ Consistent Learning
- ✓ Hands-on Practice
- ✓ Real Projects
- ✓ Never Give Up

10 Apply for Jobs

Search and apply smart.
Prepare for interviews.
Keep learning and improving.



9 Build Resume + LinkedIn

- Showcase your skills, projects and achievements.
- Create a strong online presence.



8 Get Certifications

- Earn beginner and associate level certs.
- They build trust and open doors.



7 Build Projects

- Build hands-on labs and real projects.
- Projects prove your skills.



6 Learn Security Tools

- Learn tools used in real world.
- Practice in labs and CTFs.



SIEM



IAM



Monitoring



Detection

Project Ideas:

- Cloud Monitoring Lab
- IAM Security Project
- Log Analysis Project
- Threat Detection Lab

Popular Tools:

- Splunk, Wazuh
- AWS Security Hub
- Microsoft Defender
- CloudTrail, GuardDuty

5 Learn AWS or Azure

- Learn core services and security features.
- Focus on hands-on practice.



Time:
1 - 2 Months

4 Learn Cloud Basics

- Understand what cloud is and how it works.
- Learn core cloud concepts.



Time:
2 Weeks

3 Learn Linux

- Learn Linux commands and basics.
- Practice daily - it's important!



Time:
3 Weeks

2 Learn Networking

- Understand how networks work.
- Learn IP, DNS, ports, protocols.



Time:
2 - 3 Weeks

1 Learn Computer Basics

- Start with the basics.
- Understand computers and how they work.



Time:
2 Weeks

YOU
MADE IT!
Cloud Security
Engineer



JOB ROLES YOU CAN APPLY FOR

- ✓ SOC Analyst
- ✓ Cloud Security Analyst
- ✓ Security Engineer
- ✓ Cloud Security Engineer
- ✓ Threat Analyst
- ✓ DevSecOps Engineer



You don't need to be a genius.
You only need Consistency.



Remember:

- Learn a little every day.
- Practice what you learn.
- Build. Break. Fix. Repeat.
- Stay curious and keep going!

≡ Your future in Cloud Security is waiting for you! ❤️

Real Cloud Security Project



Detect Suspicious AWS Login Activity

1. Project Goal

What does a Cloud Security Engineer do daily?

Cloud security engineers monitor cloud accounts, detect suspicious activity, investigate alerts and stop attackers.



2. Real World Scenario



Hacker tries to login from another country.



Login attempt on company AWS account.

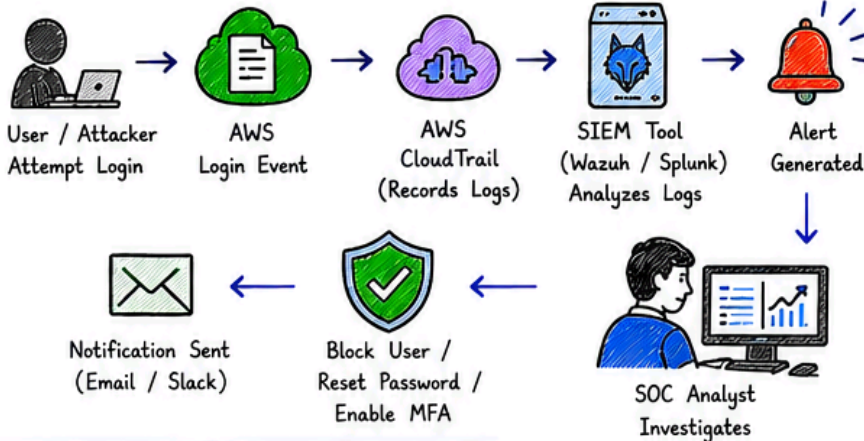


Unusual login detected! Alert is generated.



SOC Analyst investigates and takes action.

3. Project Architecture (High Level Flow)



4. Tools Used

Tool	Purpose
aws IAM	Manage users and access permissions.
CloudTrail	Records API calls and login activity.
Wazuh (or Splunk)	Detects suspicious activity and creates alerts.
MFA	Adds an extra layer of security.
Slack / Email	Sends alert notifications.

5. Daily Work Steps (Real Workflow)

- User logs into AWS (Using console or CLI)
- CloudTrail records the login event (IP, location, device, time, etc.)
- SIEM (Wazuh / Splunk) analyzes logs and detects **unusual activity**.
- Alert is generated and sent to SOC Analyst.
- SOC Analyst Investigates:
 - IP address & country
 - Login time
 - User behavior
 - Failed login attempts
- Security Team Takes Action:
 - Block the malicious IP
 - Disable / Lock the user
 - Reset password
 - Enable MFA if not enabled
 - Monitor for more activity

Example Alert

User: admin
IP: 185.199.109.153
Country: Unknown
Location: Russia
Time: 02:13 AM UTC
Result: Success



6. Skills You Will Learn

- ✓ IAM & Access Control
- ✓ Logging (CloudTrail)
- ✓ Monitoring & Alerting
- ✓ SIEM Basics (Wazuh/Splunk)
- ✓ Investigation
- ✓ Incident Response
- ✓ Cloud Security Best Practices

7. Real Job Roles



SOC Analyst



Cloud Security Analyst



Blue Team Engineer



Incident Responder

8. Beginner Practice Setup



AWS Free Tier
(Create Account)



Splunk Free
(SIEM)



Wazuh
(Open Source)



Virtual Machines
(For testing)

What you will do:

- Create IAM user
- Enable CloudTrail
- Generate login events
- Analyze in SIEM
- Create detection rule
- Get alerts & respond!

9. Key Takeaway



Cloud Security is not only theory. It is **monitoring, detecting and protecting** systems every day.

Your job = Stop attackers, protect data and keep **business** running safe!



Keep Learning.
Keep Building.
Keep Protecting.



This project is simple, real-world and perfect for beginners. Practice it, build confidence and start your cloud security career!



Ready to Start Your Cloud Security Journey?



The cloud is the future.
Security is the superpower.
You can do this! 😊



TAKE ACTION TODAY!

1 Learn



Keep learning
one concept
every day.

Small steps,
Big progress!

2 Practice



Build labs,
try tools and
work on projects.

Practice makes
you stronger!

3 Build



Build your skills,
projects and
portfolio.

Show your
work to the world!

4 Connect



Join communities,
ask questions
and help others.

Together we
grow!

5 Apply



Apply for jobs,
internships and
opportunities.

Your dream job
is waiting!



Your Goal:

Become a skilled Cloud Security Professional
and protect the cloud, data and users.



REMEMBER:



Be Patient.
Good things
take time.



Be Consistent.
Every day
matters.



Be Curious.
Ask questions,
explore more.



Never Give Up.
Every expert
was once a
beginner!



The best time to start was yesterday.
The next best time is NOW!



Let's build a secure cloud together! ❤️

Start Learning. Start Building. Start Your Journey!